Subtractive Mixture Models via Squaring: Representation and Learning

Paper Code

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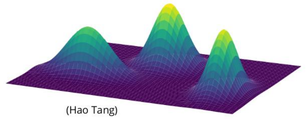
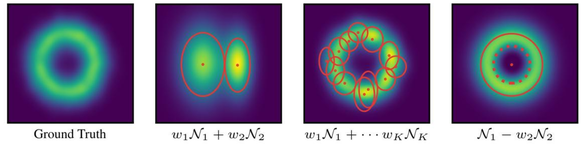
0 Mixture models

$$p(\mathbf{X}) = \sum_{i=1}^{K} w_i \, p_i(\mathbf{X}) \quad \text{subject to} \quad w_i \geq 0 \quad \sum_{i=1}^{K} w_i = 1$$



(Hao Tang)

✗ components can only be added together!



Fewer components with subtractions

Questions? ***...Contributions!*** 1 2 3

1 How to learn subtractive mixture models?

$$p(\mathbf{X}) = \sum_{i=1}^{K} w_i \, p_i(\mathbf{X}) \qquad w_i \in \mathbb{R}$$

How to ensure $p(\mathbf{X})$ is non-negative?

$\Longrightarrow$ Impose ad-hoc constraints over the parameters

✗ challenging to derive in closed-form [1][2][3]

2 How much more expressive are they?

with respect to traditional additive-only mixtures

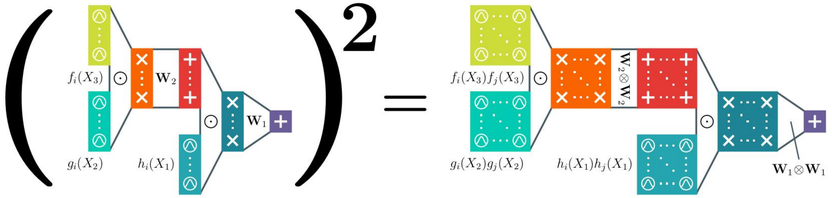
3 What is their relationship with other models?

understanding why they are expressive ...

... and why they support tractable inference

TL; DR *"We learn exponentially more expressive mixture models with subtractions, by squaring deep tensorized mixtures"*

Learning deep subtractive mixtures by squaring layers of a deep circuit

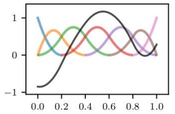


1 Squaring mixtures ...

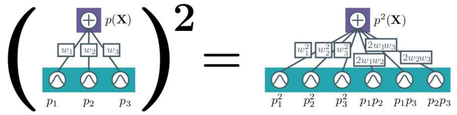
$$p(\mathbf{X}) \propto \left(\sum_{i=1}^{K} w_i \, p_i(\mathbf{X}) \right)^2 = \sum_{i=1}^{K} \sum_{j=1}^{K} w_i w_j \, p_i(\mathbf{X}) p_j(\mathbf{X})$$

Renormalization:

$$Z = \sum_{i=1}^{K} \sum_{j=1}^{K} w_i w_j \int p_i(\mathbf{X}) p_j(\mathbf{X}) \mathrm{d}\mathbf{X}$$



Tractable marginalization is supported by exponential families [2] and splines components



... by squaring circuits

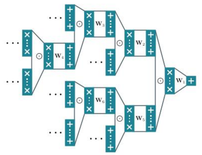
Build deep mixtures with layers as "Lego blocks"

2

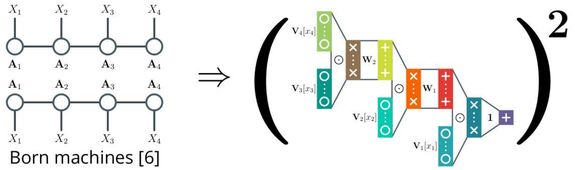
Theorem. exponential separation [4] [5]

There is a class of distributions $\mathcal{F}$ over variables $\mathbf{X}$ that can be compactly represented as a shallow squared mixture with negative weights, but the smallest structured decomposable additive-only mixture of any depth computing any $F \in \mathcal{F}$ has size $2^{\Omega(|\mathbf{X}|)}$.

Deep additive-only mixtures



Squared subtractive mixture model



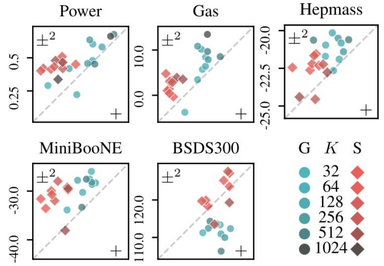
Born machines [6]

$$p(\mathbf{x}) \propto \kappa(\mathbf{x})^\top \mathbf{A} \, \kappa(\mathbf{x}) \Rightarrow \sum \left(\mathbf{W} \right)^2$$

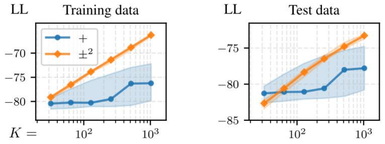
PSD models [7]

3 Understanding the expressiveness of other models in a unifying framework

Density estimation



GPT2 distillation [8]



References

[1] B. Zhang and C. Zhang. "Finite mixture models with negative components". In: *MLDM*. Springer. 2005, pp. 31–41.
[2] G. Rabusseau and F. Denis. "Learning negative mixture models by tensor decompositions". In: *arXiv preprint arXiv:1403.4224* (2014).
[3] R. Jiang, M. J. Zuo, and H. Li. "Weibull and inverse Weibull mixture models allowing negative weights". In: *Reliability Engineering & System Safety* 66.3 (1999), pp. 227–234.
[4] J. Martens and V. Medabalimi. "On the expressive efficiency of sum product networks". In: *arXiv preprint arXiv:1411.7717* (2014).
[5] A. de Colnet and S. Mengel. "A Compilation of Succinctness Results for Arithmetic Circuits". In: *KR*. 2021, pp. 205–215.
[6] I. Glasser et al. "Expressive power of tensor-network factorizations for probabilistic modeling". In: *NeurIPS*. Curran Associates, Inc., 2019, pp. 1498–1510.
[7] A. Rudi and C. Ciliberto. "PSD Representations for Effective Probability Models". In: *NeurIPS*. Curran Associates, Inc., 2021, pp. 19411–19422.
[8] H. Zhang et al. "Tractable Control for Autoregressive Language Generation". In: *ICML*. Vol. 202. Proceedings of Machine Learning Research. PMLR, 2023, pp. 40932–40945.